

IAM Task

Task: Create IAM User with EC2 and S3 Full Access

Step 1: Open IAM

1. Log in to the Amazon Web Services Management Console.
2. Search for IAM in the top search bar.
3. Click IAM.

Step 2: Create IAM User

1. In the left menu, click Users.
2. Click Create user.
3. Enter a username
4. Click Next.

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☐ Add user to group
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ Copy permissions
Copy all group memberships, attached managed policies, and inline policies from an existing user.

☒ Attach policies directly
Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (2/1584)

Choose one or more policies to attach to your new user.

Filter by Type: 1 match

☒

Policy name [↗](#)

▲

Type

▼

Attached entities

▼

☒	AmazonS3FullAccess	AWS managed	0
-------------------------------------	------------------------------------	-------------	---

☒ [Create policy](#)

► Set permissions boundary - optional

IAM > IAM users > Create user

Step 1

Specify user details

Step 2

Set permissions

Step 3

Review and create

Review and create

Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name

devops-user

Console password type

None

Require password reset

No

Permissions summary

Name	Type	Used as
AmazonEC2FullAccess	AWS managed	Permissions policy
AmazonS3FullAccess	AWS managed	Permissions policy

Tags - optional

Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags.

Cancel

Previous

Create user

User created successfully

You can view and download the user's password and email instructions for signing in to the AWS Management Console.

View user

IAM users (1) Info

Delete user

Create user

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Search

User name

Path

Group

Last activity

MFA

Password age

Console last sign-in

Access key ID

devops-user

/

0

-

-

-

-

-

IAM user sign in

Account ID or alias (Don't have?)

814787354428

☐ Remember this account

IAM username

devops-user

Password

☐ Show Password [Having trouble?](#)

Sign in

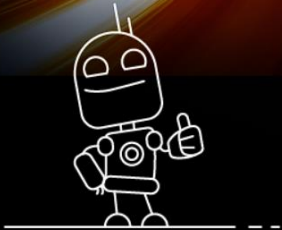
Sign in using root user email

Create a new AWS account

Amazon Lightsail

Lightsail is the easiest way to get started on AWS

Learn more »



Task 2: Create IAM Group with EC2 Read Access

Step 1: Open IAM

1. Log in to the AWS Management Console.
2. Search for IAM.
3. Click IAM.
4. In the left-side menu, click User groups.

Step 2: Create a Group

1. Click Create group.
2. Enter a group name

Step 3: Create the Group

1. Check that the policy is selected:
2. AmazonEC2ReadOnlyAccess
3. Click Create group.
4. Your group is now created.

IAM user groups (1/1) [Info](#)

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users.

☒

Group name

▲

☐

Users

▼

☐

Permissions

▼

☐

Creation time

▼

☒	EC2-ReadOnly-Group	0	Defined	1 minute ago
-------------------------------------	------------------------------------	---	---------	--------------

EC2-ReadOnly-Group [Info](#)

Delete

Summary

Edit

User group name

EC2-ReadOnly-Group

Creation time

October 06, 2026, 22:10 (UTC+05:30)

ARN

arn:aws:iam::814787354428:group/EC2-ReadOnly-Group

Users

Permissions

Access Advisor

Permissions policies (1) [Info](#)

You can attach up to 10 managed policies.

Filter by Type

All types

▼

☐

Policy name [↗](#)

▲

☐

Type

▼

☐

Attached entities

▼

☐	AgentRegistryReadOnlyAccess	AWS managed	1
--------------------------	---	-------------	---

3. Create a new user named "Devops" and add to the group created in task 2.

Step 1: Open IAM

1. Log in to AWS Management Console.
2. Search for IAM.
3. Open IAM.
4. From the left menu, click Users.

Step 2: Create the User

1. Click Create user.
2. Enter the username:
3. Click Next.

Step 3: Add User to Existing Group

On the Permissions page:

1. Select Add user to group.
2. You will see the groups available in your account.
3. Select the group you created in Task 2
4. Click Next.

IAM users (1/2) [Info](#)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

☐	User name	Path	Group	Last activity	MFA	Password age	Console last sign-in	Access key ID
☒	Devops	/	0	-	-	Now	-	-
☐	devops-user	/	0	-	-	28 minutes	-	-

Devops [Info](#) [Delete](#)

Summary

ARN
arn:aws:iam::814787354428:user/Devops

Created
October 06, 2026, 22:21 (UTC+05:30)

Console access
Enabled without MFA

Last console sign-in
Never

Access key 1
[Create access key](#)

Permissions [Groups \(1\)](#) [Tags](#) [Security credentials](#) [Last Accessed](#)

Permissions policies (1) [Simulate](#) [Remove](#) [Add permissions](#)

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type [All types](#)

☐	Policy name	Type	Attached via
☐	AgentRegistryReadOnlyAccess	AWS managed	Group EC2-ReadOnly-Group

► **Permissions boundary (not set)**

4. Write a bash script to create an IAM user with VPC full access.

```

~/m/'
[ec2-user@ip-10-0-0-116 ~]$ aws --version
aws-cli/2.36.47 Python/3.9.25 Linux/6.18.51-120.163.amzn2023.x86_64 source/x86_64.amzn.2023
[ec2-user@ip-10-0-0-116 ~]$

```

```
[root@ip-10-0-0-116 ~]# vi create-vpc-user.sh
[root@ip-10-0-0-116 ~]# cat create-vpc-user.sh
#!/bin/bash

# IAM username
USERNAME="VPC-Admin"

# Create IAM user
aws iam create-user --user-name "$USERNAME"

# Attach VPC full access policy
aws iam attach-user-policy \
  --user-name "$USERNAME" \
  --policy-arn arn:aws:iam::aws:policy/AmazonVPCFullAccess

echo "IAM user $USERNAME created successfully."
echo "VPC Full Access policy attached."

[root@ip-10-0-0-116 ~]#
```

1 policy added

Devops [Info](#) [Delete](#)

Summary

ARN
arn:aws:iam::814787354428:user/Devops

Created
October 06, 2026, 22:21 (UTC+05:30)

Console access
Enabled without MFA

Last console sign-in
Never

Access key 1
[Create access key](#)

Permissions [Groups \(1\)](#) [Tags](#) [Security credentials](#) [Last Accessed](#)

Permissions policies (2) [Refresh](#) [Simulate](#) [Remove](#) [Add permissions](#)

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type
All types

☐	Policy name	Type	Attached via
☐	AgentRegistryReadOnlyAccess	AWS managed	Group EC2-ReadOnly-Group
☐	AmazonVPCFullAccess	AWS managed	Directly

► **Permissions boundary** (not set)

5. Create an IAM policy to allow EC2 access for a specific user in specific regions only

Step 1: Open IAM

1. Open AWS Console.
2. Search for IAM.
3. Click IAM.
4. On the left side, click Policies.
5. Click Create policy.

Step 2: Choose JSON

1. At the top, select the JSON tab.
2. Delete the existing JSON and enter.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "DenyEC2OutsideMumbai",
      "Effect": "Deny",
      "Action": "ec2:*",
      "Resource": "*",
      "Condition": {
        "StringNotEquals": {
          "aws:RequestedRegion": "ap-south-1"
        }
      }
    }
  ]
}
```

Step 3: Attach the policy to the user

Go to:

IAM → Users → Select your user → Permissions

Click:

Add permissions → Attach policies directly

Step 4: Name the policy

Click Next or continue to the policy details.

Enter a name such as: Devops

Description:

Allow EC2 access only in Mumbai region.

Click Create policy.

Specify permissions Info

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Policy editor

```
1
2 {
3   "Version": "2012-10-17",
4   "Statement": [
5     {
6       "Sid": "DenyEC2OutsideMumbai",
7       "Effect": "Deny",
8       "Action": "ec2:*",
9       "Resource": "*",
10      "Condition": {
11        "StringNotEquals": {
12          "aws:RequestedRegion": "ap-south-1"
13        }
14      }
15    }
16  ]
17 }
18
19
20
```

Policy Ec2RegionSpecific created.

[View policy](#)

Policies (1586) Info

⌕ Ec2Reg

Filter by Type

All types

1 match

Policy name	Type	Used as
Ec2RegionSpecific	Customer managed	None

IAM > IAM users > Devops > Add permissions

Step 1

Add permissions

Step 2

Review

Review

The following policies will be attached to this user. [Learn more](#)

User details

User name
Devops

Permissions summary (1)

Name	Type	Used as
Ec2RegionSpecific	Customer managed	Permissions policy

Cancel

Previous

Add permissions

1 policy added

Devops

Info

Delete

Summary

ARN
arn:aws:iam::814787354428:user/Devops

Created
October 06, 2026, 22:21 (UTC+05:30)

Console access
Enabled without MFA

Last console sign-in
Never

Access key 1
Create access key

Permissions

Groups (1)

Tags

Security credentials

Last Accessed

Permissions policies (3)

Permissions are defined by policies attached to the user directly or through groups.

Search

Filter by Type
All types

< 1 >

☐	Policy name	Type	Attached via
☐	AgentRegistryReadOnlyAccess	AWS managed	Group EC2-ReadOnly-Group
☐	AmazonVPCFullAccess	AWS managed	Directly
☐	Ec2RegionSpecific	Customer managed	Directly

6. We have two accounts: Account A and Account B. Account A user should access an S3 bucket in Account B.

Devops

Info

Delete

Summary

ARN
arn:aws:iam::814787354428:user/Devops

Created
October 06, 2026, 22:21 (UTC+05:30)

Console access
Enabled without MFA

Last console sign-in
Never

Access key 1
Create access key

Permissions

Groups (1)

Tags

Security credentials

Last Accessed

Permissions policies (3)

Permissions are defined by policies attached to the user directly or through groups.

Search

Filter by Type
All types

< 1 >

☐	Policy name	Type	Attached via
☐	AgentRegistryReadOnlyAccess	AWS managed	Group EC2-ReadOnly-Group
☐	AmazonVPCFullAccess	AWS managed	Directly

Instance summary for i-099acf9b776261ee0 (bash-script) Info

Updated 1 minute ago

Instance ID

i-099acf9b776261ee0

IPv6 address

—

Hostname type

IP name: ip-10-0-0-116.eu-north-1.compute.internal

Answer private resource DNS name

—

Auto-assigned IP address

13.61.0.233 [Public IP]

IAM role

—

IMDSv2

Required

Operator

—

Public IPv4 address

13.61.0.233 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-10-0-0-116.eu-north-1.compute.internal

Instance type

t3.micro

VPC ID

vpc-078c669fe95b08aae (bash-scripting)

Subnet ID

subnet-07a07764ff2660b7f (bash-subnet)

Instance ARN

arn:aws:ec2:eu-north-1:814787354428:instance/i-099acf9b776261ee0

Private IPv4 addresses

10.0.0.116

Public DNS

—

Elastic IP addresses

—

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

—

Managed

false

Details

Status and alarms

Monitoring

Security

Networking

Storage

Taags

Amazon S3 > Buckets > account-b-cross-access-2026

Successfully created bucket "account-b-cross-access-2026"

account-b-cross-access-2026 Info

Objects

Metadata

Properties

Permissions

Metrics

Management

File systems

Access Points

Objects (0)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Find objects by prefix

Name

Type

Last modified

Size

Storage class

No objects

You don't have any objects in this bucket.

Upload

Edit bucket policy Info

Bucket policy

Policy examples

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. Learn more

Bucket ARN

arn:aws:s3:::account-b-cross-access-2026

Policy

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Principal": {
7         "AWS": "arn:aws:iam::814787354428:user/S3-CrossAccount-User"
8       },
9       "Action": "s3:ListBucket",
10      "Resource": "arn:aws:s3:::account-b-cross-access-2026"
11    },
12    {
13      "Effect": "Allow",
14      "Principal": {
15        "AWS": "arn:aws:iam::814787354428:user/S3-CrossAccount-User"
16      },
17      "Action": [
18        "s3:GetObject",
19        "s3:PutObject",
20        "s3:DeleteObject"
21      ],
22      "Resource": "arn:aws:s3:::account-b-cross-access-2026/*"
23    }
24  ]
25 }
```

Edit statement

Remove

Add actions

Choose a service

Search services

Included

S3

Available

AI Operations

AMP

API Gateway

API Gateway V2